Q.24 Draw the cam profile for a radial knife edge follower having following conditions: (CO1)

Lift = 50 mm;

Base circle radius = 50 mm;

Rise: 60° of cam rotation with SHM; then

Dwell for 45° of cam rotation ; then

Fall: 90° of cam rotation in SHM; then

Dwell for remaining period.

Q.25 Draw front view of a four-cylinder engine crankshaft. (CO1,CO2)

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220332/170333/120333 /030333/030345/344

3rd Sem / Automobile

Subject : Auto Engineering Drawing / Automobile Engineering Drawing / Auto Engineering Drawing - I / Auto Engineering Drawing - II

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The differences between the two limits of size of the part is called (CO3)

a) Tolerance b) Low limit

c) High limit d) Design size

Q.2 What does '50' represents in 50H8/g7? (CO3)

a) Basic size

b) Actual size

c) Maximum limit of size

d) Minimum limit of size

Q.3 A tapered roller bearings is able to carry: (CO2)

a) Radial loads only

b) Axial loads only

c) Both radial and axial loads

d) Only loads perpendicular to the shaft

Q.4 The primary function of a leaf spring in a vehicle is (CO2)

a) To transmit engine power to the wheels
b) To absorb shocks and vibrations from the road
c) To steer the vehicle
d) To brake the vehicle

Q.5 The function of master cylinder in hydraulic brakes is to (CO2)

a) builds up hydraulic pressure to operate the brakes
b) maintains constant volume of fluid in the system
c) serves as a pump to force air out of the hydraulic system
d) All of the above

Q.6 The small end of the connecting rod is connected to the: (CO2)

a) Crankshaft b) Piston
c) Cylinder block d) Oil pump

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define shaft basis system. (CO3)

Q.8 Write one difference between ball bearing & roller bearing. (CO2)

Q.9 Define module of Gear.

Q.10 Define addendum of a gear. (CO2)

Q.11 Valve mechanism is operated by________ (CO2)

Q.12 Draw roller type of cam follower. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Show bilateral tolerance with the help of a diagram. (CO3)

Q.14 Explain shaft basis system. (CO3)

Q.15 Draw a sketch of bush bearing. (CO1)

Q.16 Name any four components of a universal joint. (CO2)

Q.17 Draw freehand sketch of a wheel cylinder. (CO1)

Q.18 Draw freehand sketch of side valve mechanism. (CO1)

Q.19 Name any four parts of a diesel engine piston. (CO2)

Q.20 Show any two types of cams with the help of diagrams. (CO1)

Q.21 An involute teeth gear is having 18 teeth, pitch circle diameter 150 mm and pressure angle 20°. Calculate its addendum and Dedendum. (CO2)

Q.22 Draw the free hand sketch of Leaf spring. (CO1)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw a profile of involute teeth by Prof. unwin's method having 22 teeth, module pitch 10 mm and pressure angle 22°. Draw at least three teeth.